

The background of the slide is a collage of several smartphone screens displaying the HYPE app interface. The screens are tilted at various angles, creating a dynamic, layered effect. The app's interface includes a top status bar with the time and battery level. Below this, there's a header with the HYPE logo and a balance of 2.225,00 €. The main content area features several interactive elements: a blue button labeled 'Invita amici', a 'Radar' section with a '1.000,00€' balance, a 'Sfrutta al meglio' section, and a 'Vai a Radar' button. On the right side, there's a list of services: 'Bonifico', 'Bollettino', 'Paga in negozio', 'F24 semplificato', and 'Ricarica telefonica'. At the bottom, there's a 'Carta' section showing a 'CARTA HYPE' with a 'SCADENZA 12/28' and a 'Mostra dati' button. The overall design is clean and modern, with a focus on financial management and digital services.

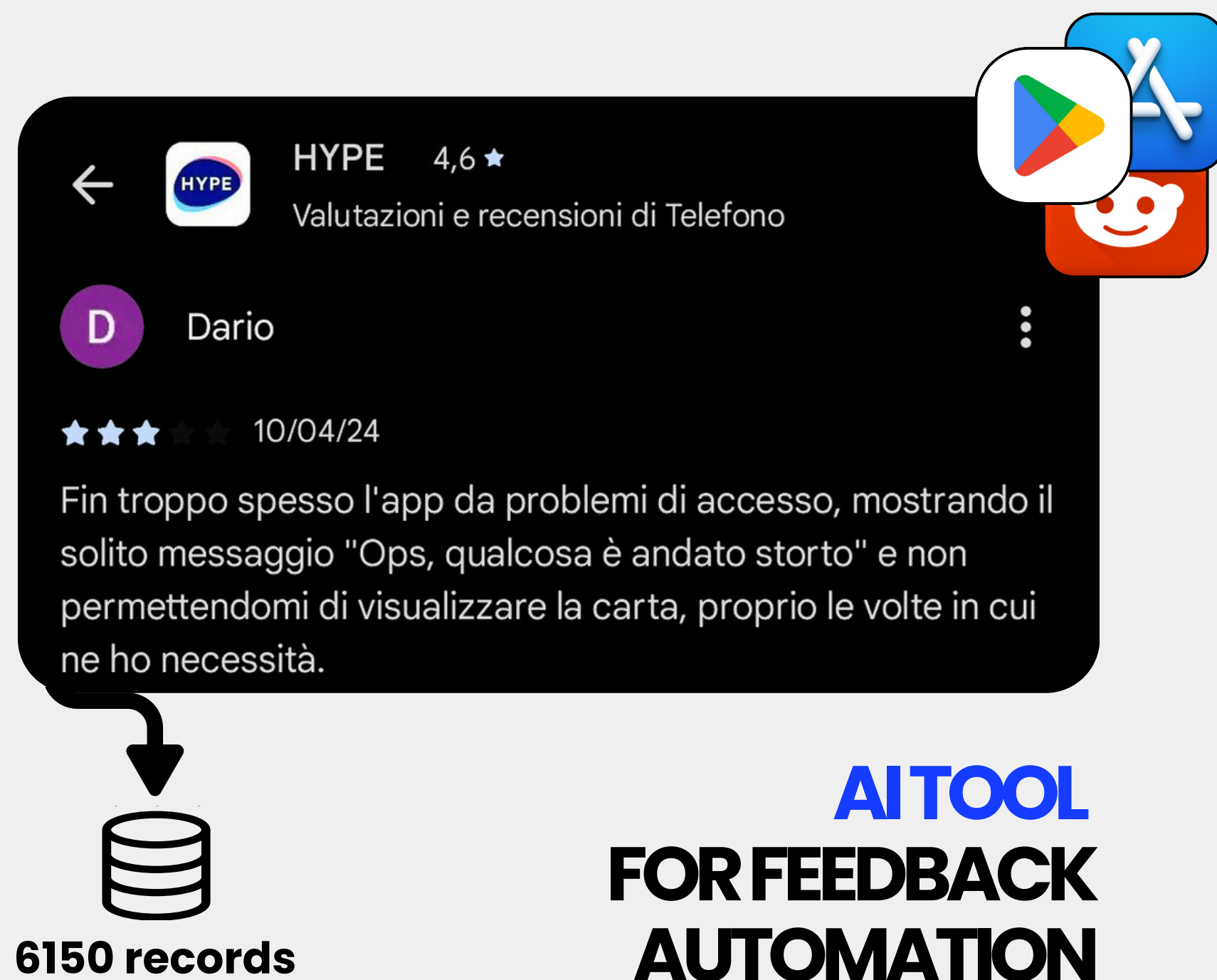
TOPIC DETECTION in App Reviews

A Data-Driven Solution for HYPE



Alessia Ciccaglione - s344499
Amirhossein Lotf Ranaei - s323336
Sonia Foco - s343043

OBJECTIVES of Our Solution:



KEY FEATURES

Automatically analyzes every HYPE app review

Detects **negative** topics (bugs, complaints, usability issues)

Identifies **recurring problems** and emerging trends

Tags each review **consistently**

BENEFITS

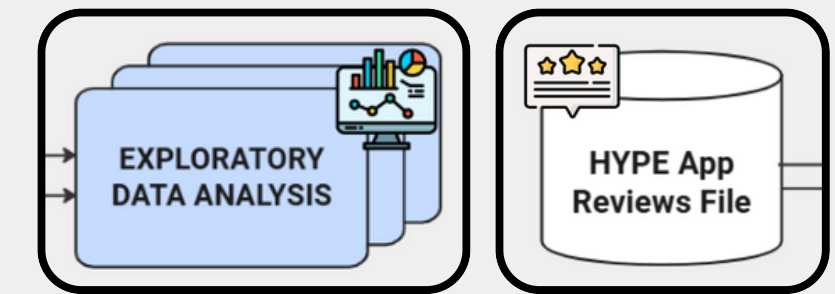
Faster insights for the Research & Customer Insight team

Reduced manual effort

Summarizes large volumes of feedback

Improved visibility on customer **pain points**

Introducing the DATASET



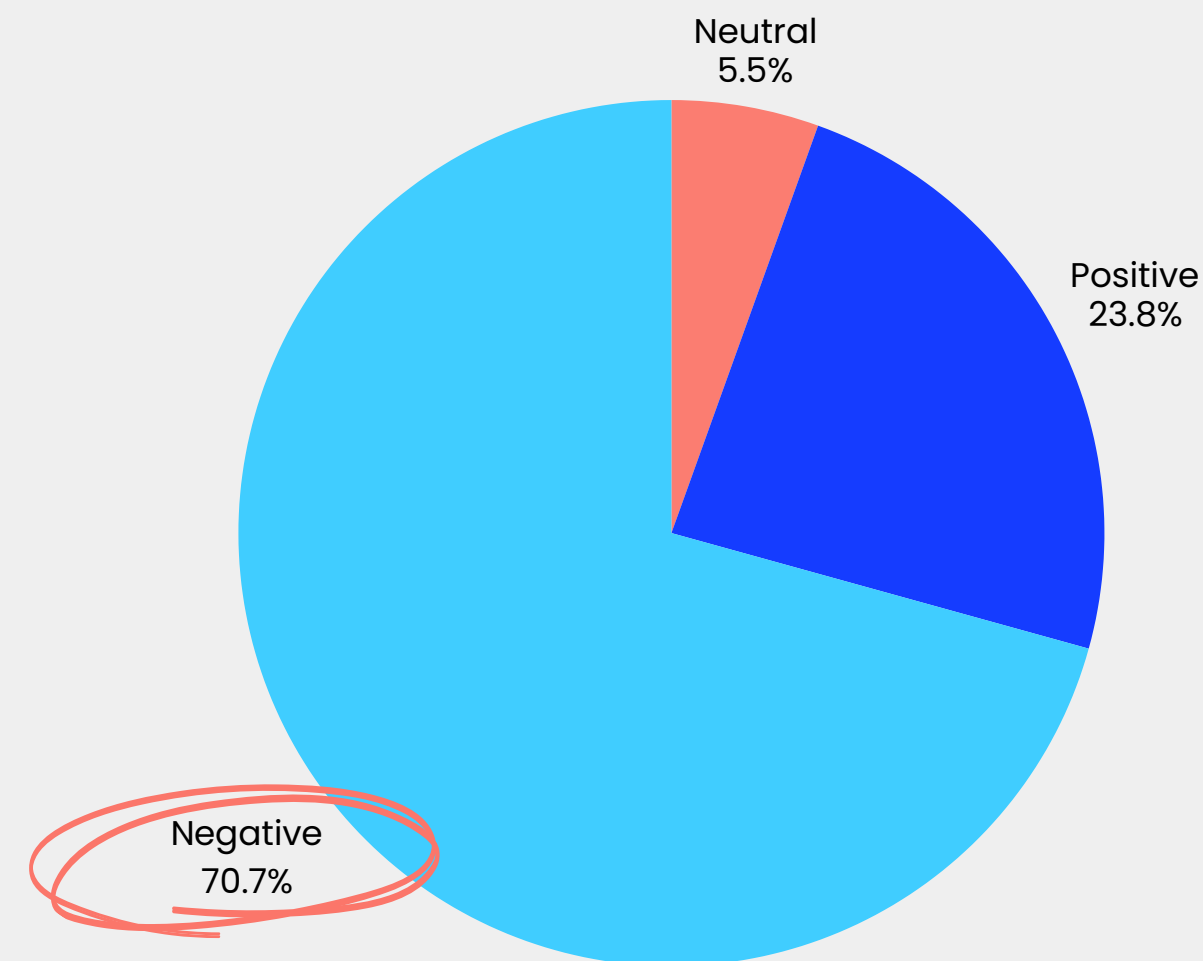
id_review	review	sentiment	rating	labels	is_labels	model_version	review_date
00000038-3b3b-5102-be9a-7cf800000000	Nell'applicazione (ricarica conto) non ho più la possibilità di ricaricare in contanti non esiste più l'opzione è cambiato qualcosa?	Negative	5	["problemi_ricarica"]	FALSO	1.0.0	01/10/2024

REAL USER REVIEWS

Ambiguous meaning
Multi-label reviews
Multilingual content
Off-topic reviews
Copy-pasted content
Spam and low-quality reviews



Small size **Labelling errors**



Defining the **VALUE PROPOSITION**

8 DECENT WORK AND ECONOMIC GROWTH



By using automated NLP, we enhance feedback analysis **efficiency**, reduce **manual effort**, and **accelerate** decision-making. Real-time insights from user reviews enable data-driven product improvements, **driving growth** and fostering a **more sustainable** business environment.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

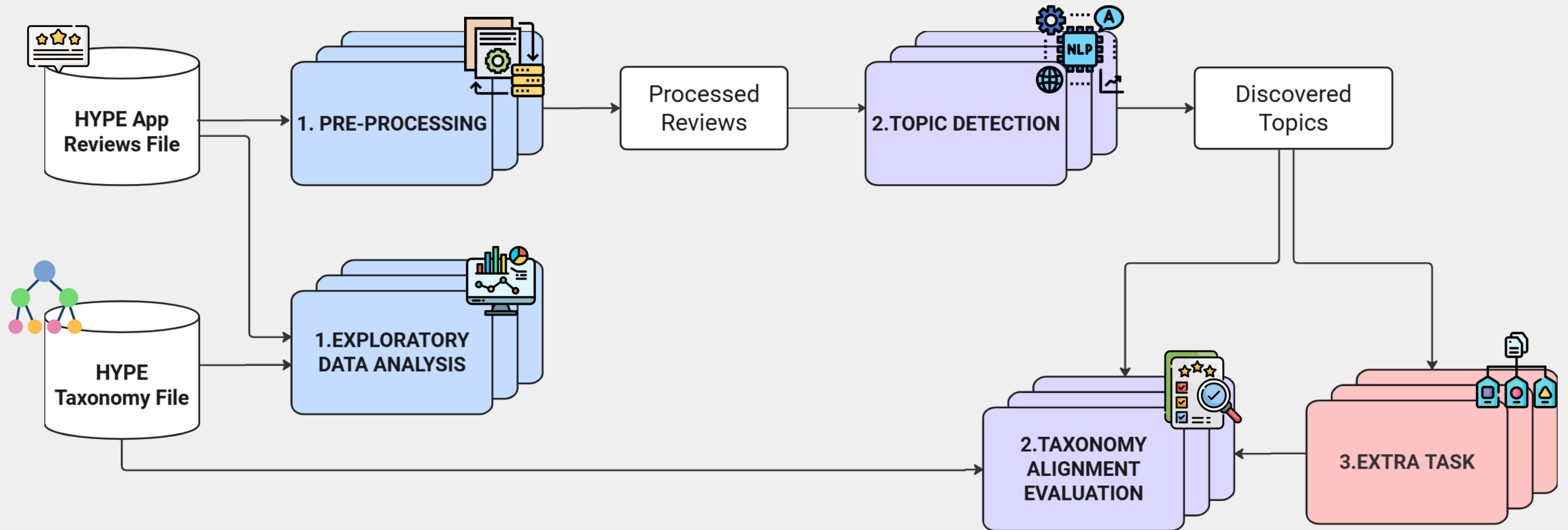


By utilizing cutting-edge **AI technology**, we automate topic detection, improving **operational efficiency** and modernizing product development processes. This supports the growth of a more **data-driven** and **technologically advanced infrastructure**.



Functional Diagram 3-STEP SYSTEM OVERVIEW

METHODOLOGY OVERVIEW



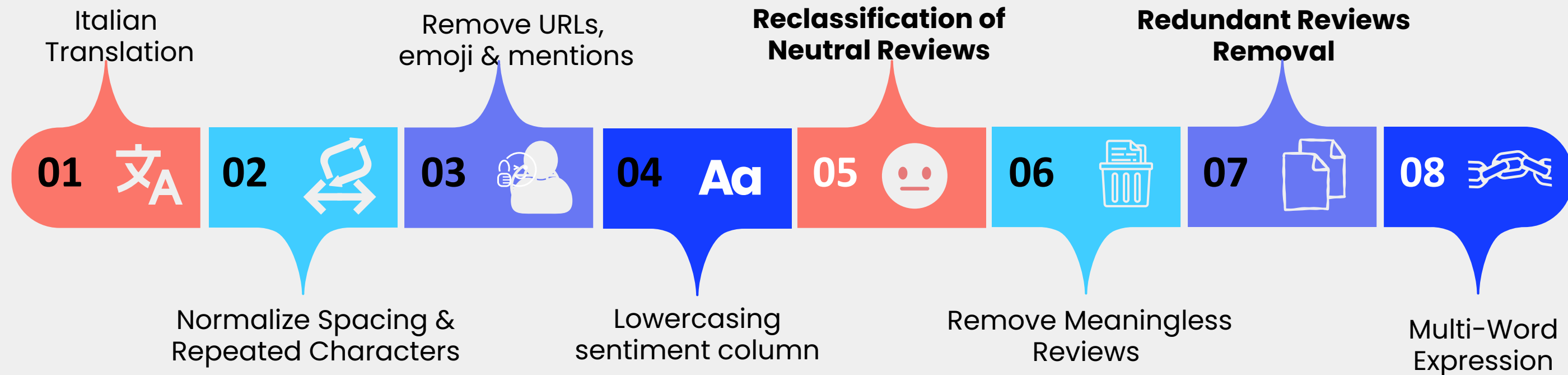
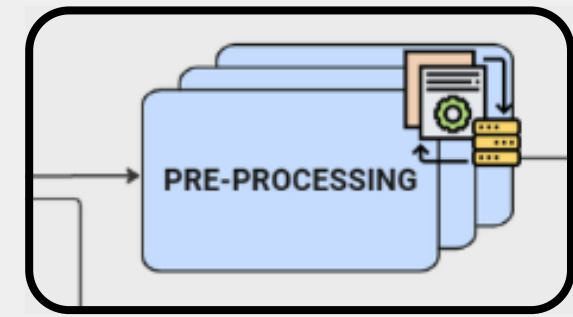
HYPOTHESIS AND RESEARCH QUESTIONS

Driving Our Investigation



- **RQ1** — Does machine translation affect the semantic quality of topic detection on multilingual app reviews?
- **RQ2** — How do different stopwords-removal strategies affect the performance of embedding-based topic modeling on app reviews?
- **RQ3** — Which configuration of BERTopic's modular components (embeddings, dimensionality reduction, clustering, and topic representation) yields the best topic quality?
- **RQ4** — Do semi-supervised topic models outperform unsupervised approaches in identifying relevant issues from app reviews?
- **RQ5** — Do embedding-based semantic similarity methods provide reliable mappings between data-driven topics and the existing taxonomy of app issues?

PRE-PROCESSING Workflow



05

Apro l'app più volte a settimana e ogni volta mi chiede di inserire la password e poi parte il Face ID.	
Vi prego fate qualcosa	Neutral
sì deve sempre migliorare	Neutral



Neutral $\neq$ always problem-free

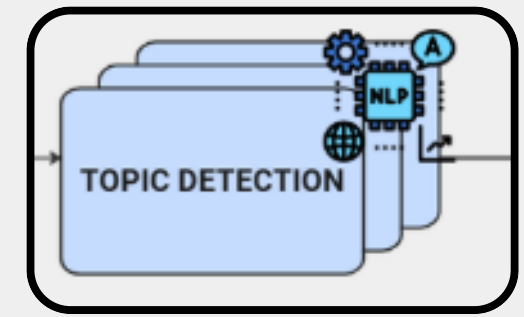
07

Review 2: Non proteggono dalle frodi, dicendo di bloccare la carta e di emetterne una nuova e a pagamento

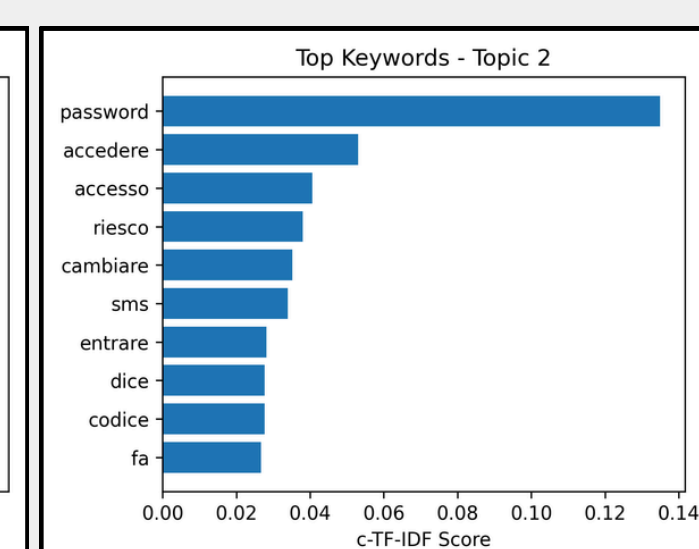
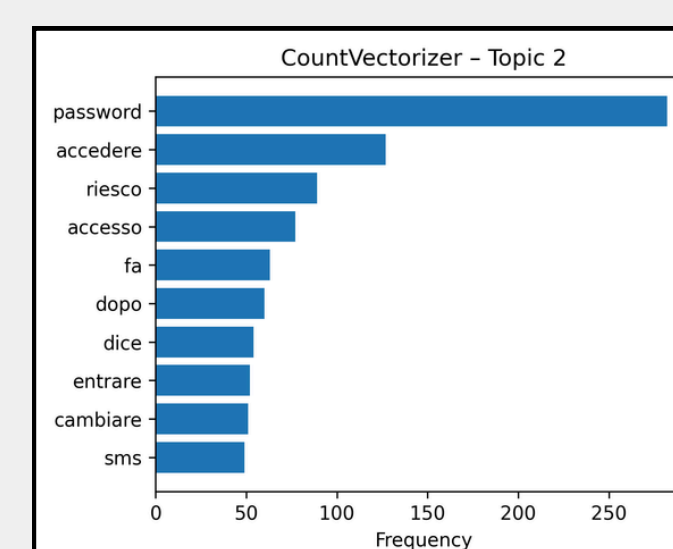
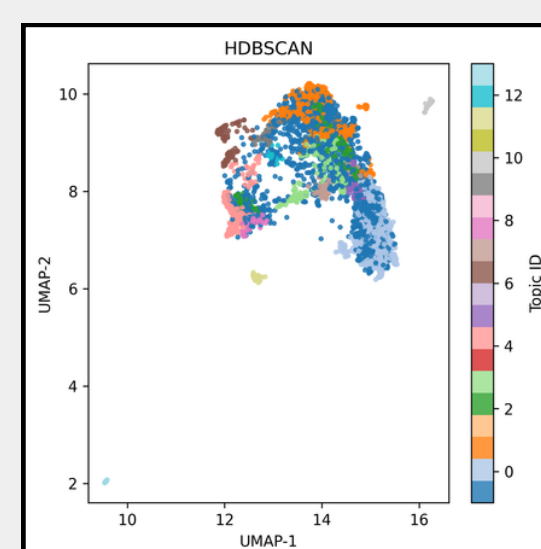
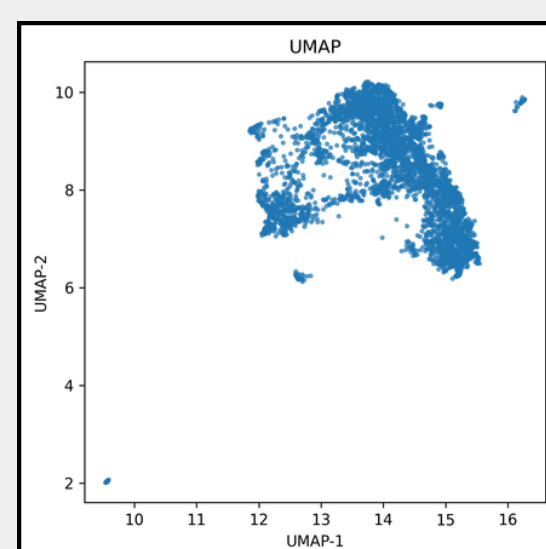
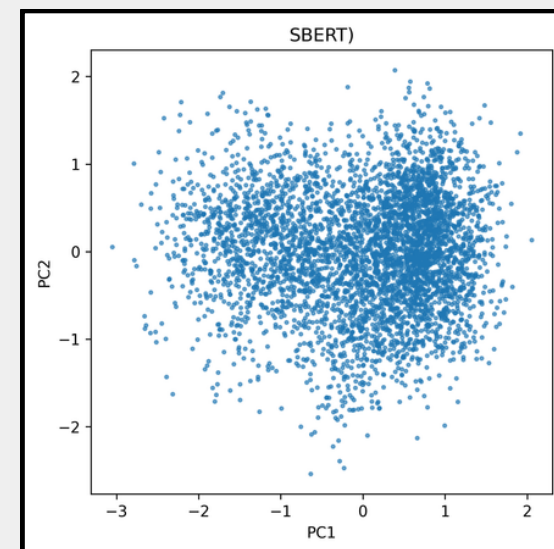
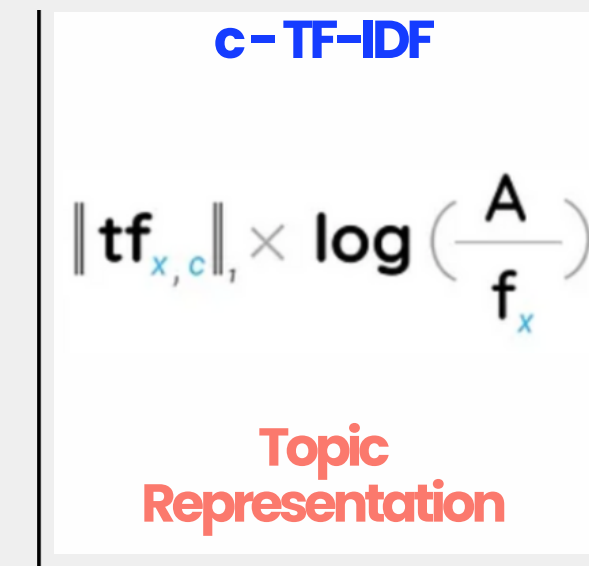
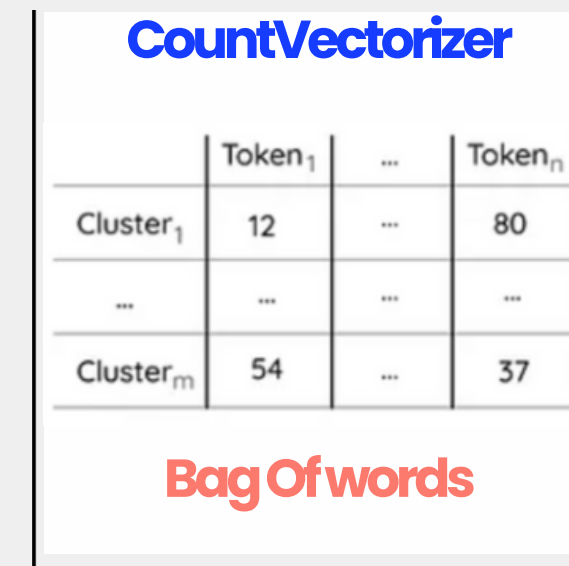
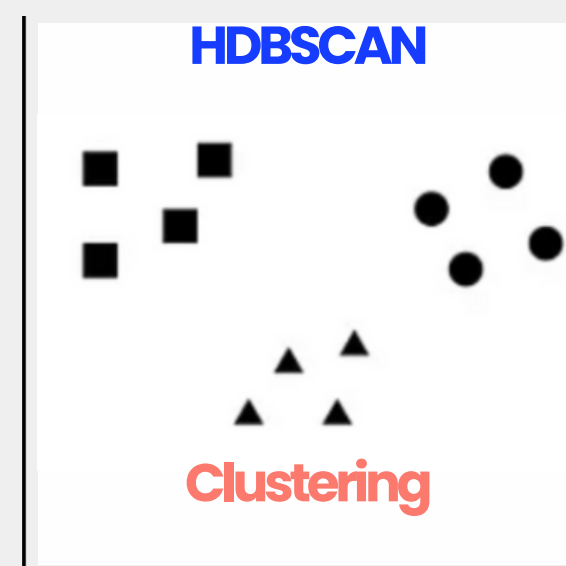
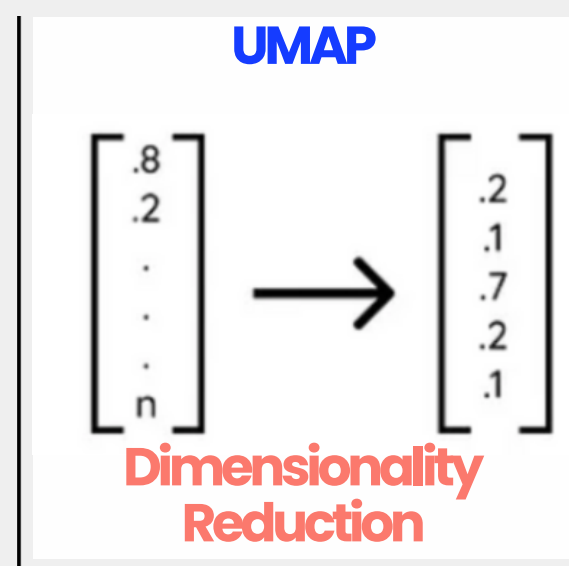
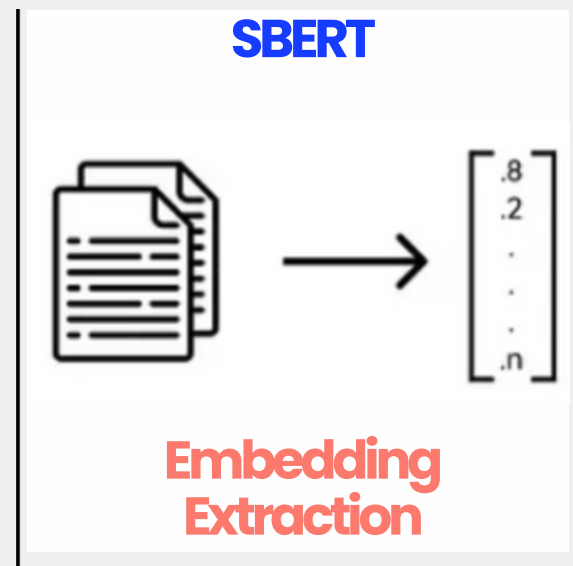
Review 1: Chiuderò il conto Hype. Non proteggono dalle frodi, dicendo di bloccare la carta e di emetterne una nuova e a pagamento

After comparing TF-IDF, SBERT, and E5, we opted for a basic **TF-IDF approach**.

Topic Detection BERTopic



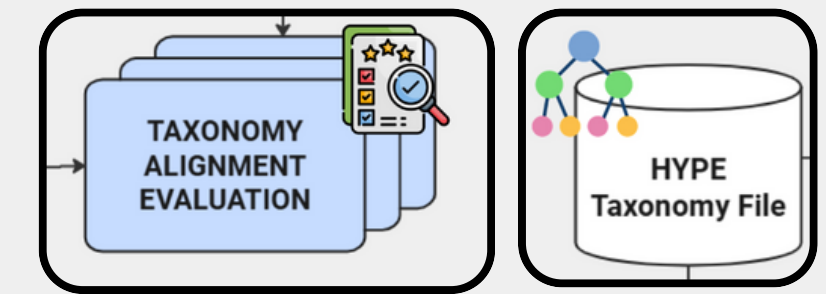
LLM-based
description
of the
topics



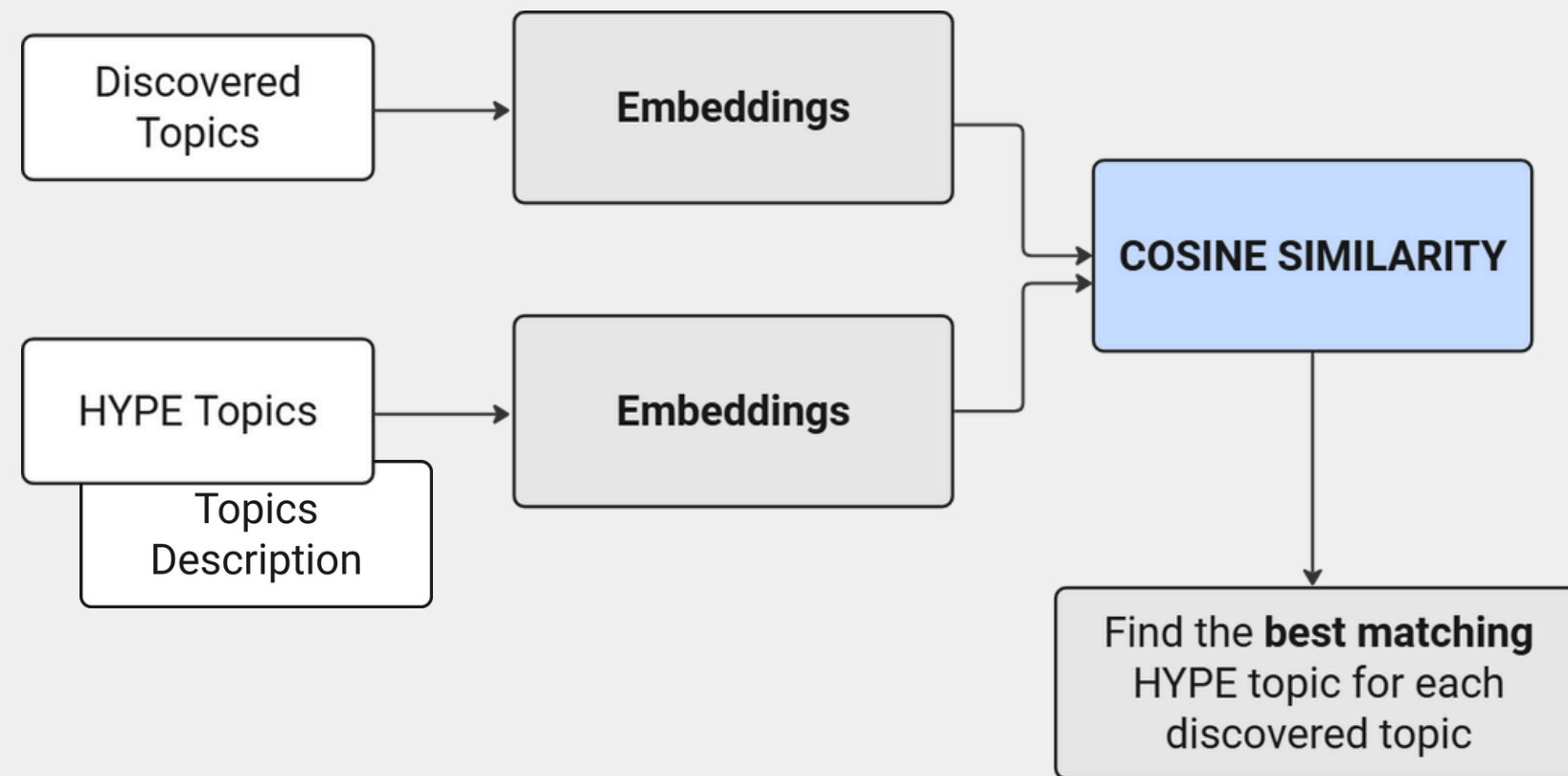
For its enterprise application, the system allows the final number of topics to be **manually defined** by the user directly or through the topic reduction function, after a qualitative evaluation of the initial BERTopic output.

"Login Issues"

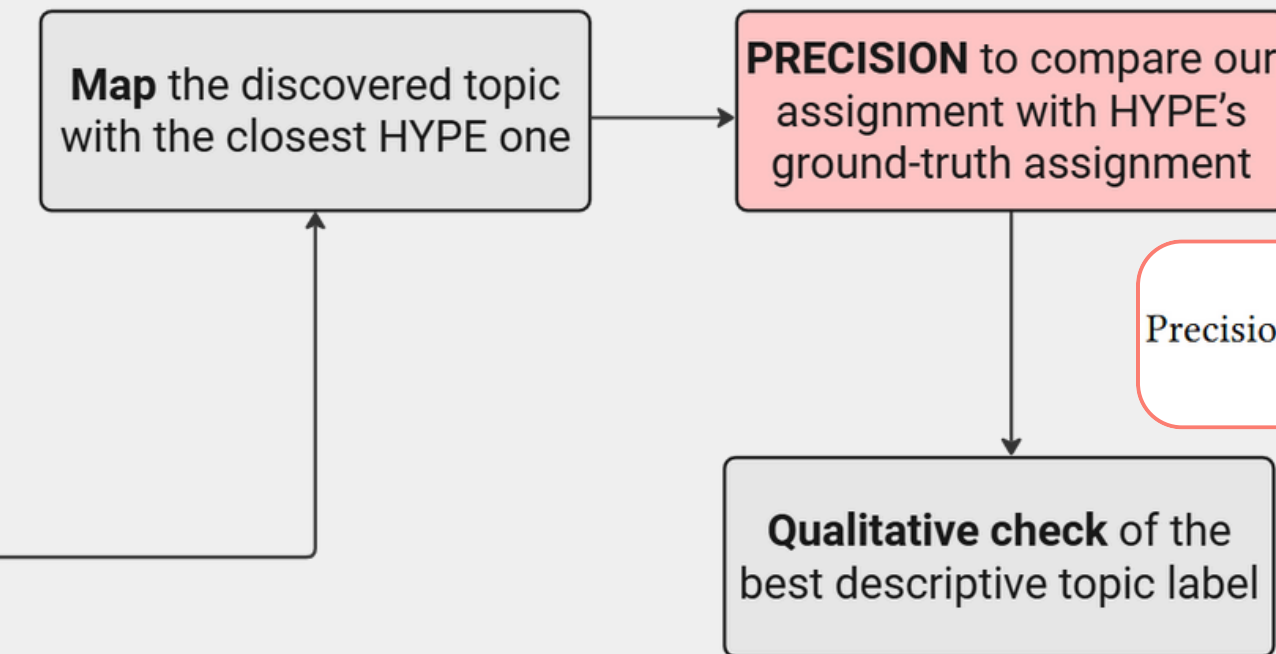
Aligning Topics to the HYPETAXONOMY



TAXONOMY ALIGNMENT



EVALUATION AND COMPARISON



$$\text{Precision}_{\text{micro}}^{(m)} = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{Y}_i^{(m)} \cap Y_i|}{|\hat{Y}_i^{(m)}|}$$

$$\text{Precision} = \frac{1}{4} (1+0+1+1) = 0.75$$

DISCOVERED_TOPIC_WORDS	BEST_MATCH_TOPIC
password accedere cambiare riesco accesso sms entrare errore codice fa	problemi_accesso
outlier	[]
chat operatore assistenza parlare bot servizio_clienti mail problema rispondono via	problemi_comunicazione
aggiornamento dopo ultimo aggiorna aggiornamenti funziona versione riesco dice continua	problemi_aggiornamento_app



HYPE_TOPIC	BEST_MATCH_TOPIC	Correct/Predicted
['problemi_accesso']	problemi_accesso	1/1 = 1
['problemi_comunicazione', 'problemi_ricarica']	problemi_accesso	0/1 = 0
['problemi_comunicazione', 'problemi_ricarica']	problemi_comunicazione	1/1 = 1
['problemi_aggiornamento_app']	problemi_aggiornamento_app	1/1 = 1

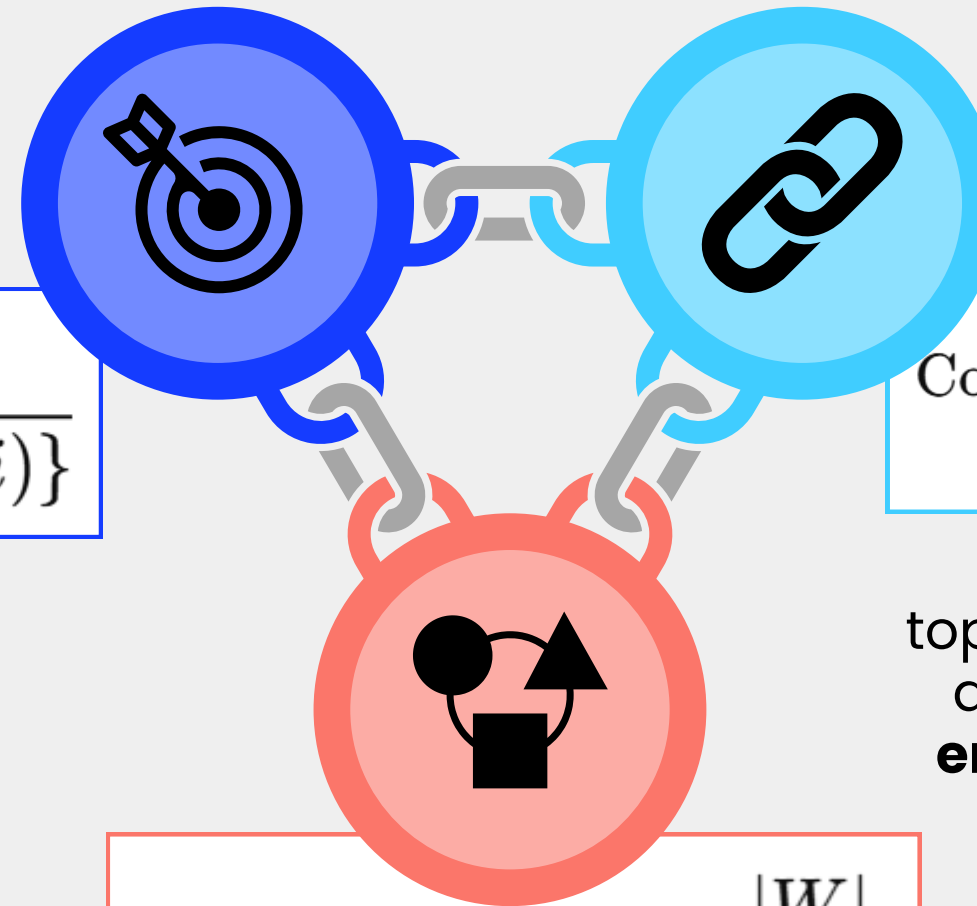
General Evaluation METRICS

EXPERIMENT & RESULT

Silhouette Score

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

Measures how well documents are clustered by comparing the average **intra-cluster distance** a of each document with the average **inter-cluster distance** b .



Topic Coherence Score

$$\text{Coherence}(t) = \frac{1}{|P|} \sum_{(w_i, w_j) \in P} \cos(\mathbf{e}_{w_i}, \mathbf{e}_{w_j})$$

Measures the semantic consistency of topics by computing, for each topic t , the average **cosine similarity between the embeddings** $\mathbf{e}_{w_i}$ and $\mathbf{e}_{w_j}$ of all pairs of its top representative words.

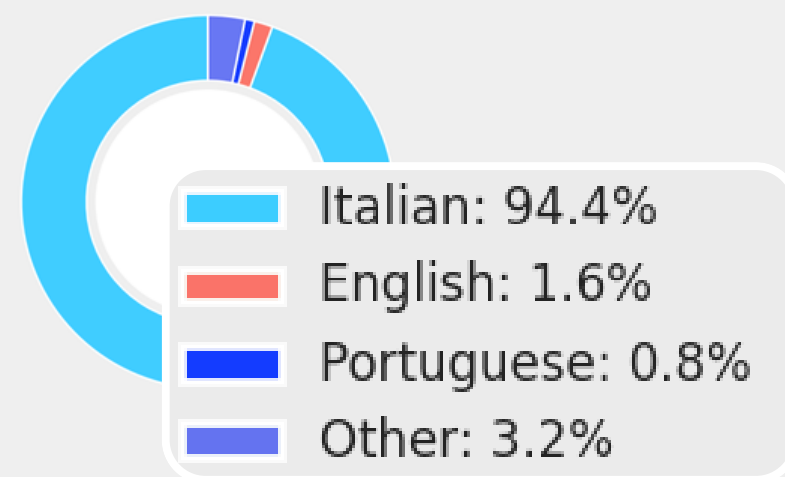
$$\text{TopicDiversity} = \frac{|W|}{T \cdot K}$$

Measures how different topics are from each other by computing the ratio between the **number of unique words** W and the total number of representative **words across all topics** T , considering the top $K=10$ words per topic.

Topic Diversity Score

Handling **MULTILINGUAL REVIEWS**

Method and Results



Some reviews are written in languages **other than Italian**. To ensure consistent analysis, all non-Italian reviews are automatically **translated into Italian** using NLLB.

 **NLLB** – No Language Left Behind

worst app 👎



l'app peggiore 👎

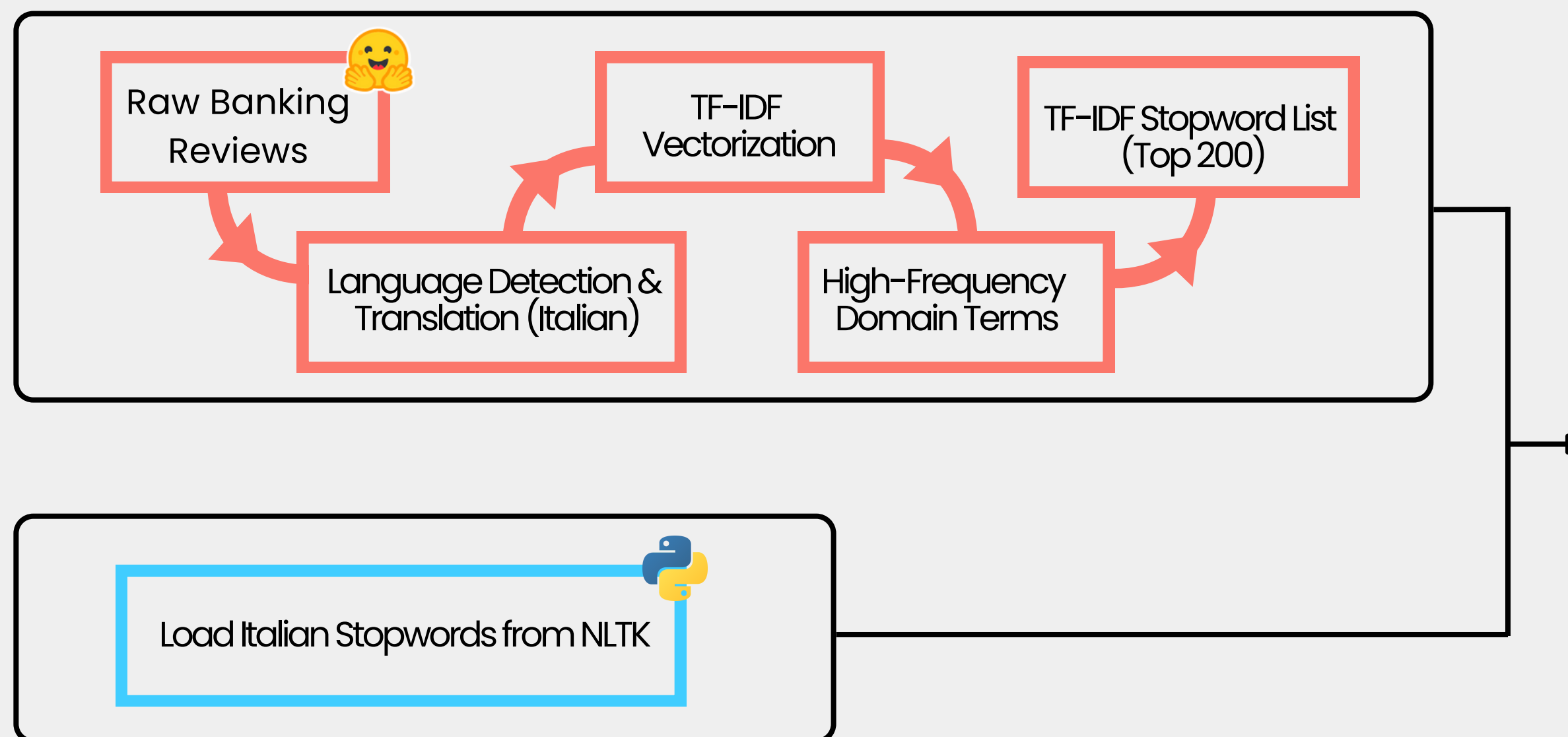
Metric	All Reviews	Translated Reviews
Silhouette Score	0.0636	0.068
Precision	0.1951	0.2031

Machine translation **does not** degrade topic semantic quality!

Precision is computed by comparing the inferred taxonomy labels with the reference labels for each review. It measures the proportion of predicted labels that correctly match the reference label set.

Comparative Evaluation of **STOPWORD STRATEGIES:** Method and Results

DOMAIN-SPECIFIC STOPWORD LIST



GENERIC STOPWORD LIST



No Stopword Removal



Generic Stopword List



**Domain-Specific
Stopword Removal**



**Delta Stopwords
(Domain – Generic)**



**Union Stopwords
(Domain + Generic)**

[Stop Words for Processing Software Engineering Documents: Do they Matter?]

Comparative Evaluation of **STOPWORD STRATEGIES:** Method and Results

Strategy	Silhouette	Topic Coherence	Topic Diversity
No Stopword Removal	0.0636	0.5375	0.8071
Generic Stopword List	0.0508	0.5313	0.8000
Domain-Specific Stopword Removal	0.0427	0.4993	0.8786
Delta Stopword (Domain - Generic)	0.0389	0.5044	0.8857
Union Stopword	0.0434	0.5236	0.8714

[Results computed using the default approach based on UMAP and DBSCAN.]

Comparing MODELS

LDA

Probabilistic model where the document is a mixture of topics, topic as a distribution over words

Topic Coherence Score: 0.6632

Topic Diversity Score: 0.3333

NMF

Matrix factorization of the document-term matrix into topic components.

Topic Coherence Score: 0.6190

Topic Diversity Score: 0.6867

BERTopic

Embedding-based model that uses transformer representations, HDBSCAN clustering and c-TF-IDF keyword extraction.

Topic Coherence Score: 0.5393

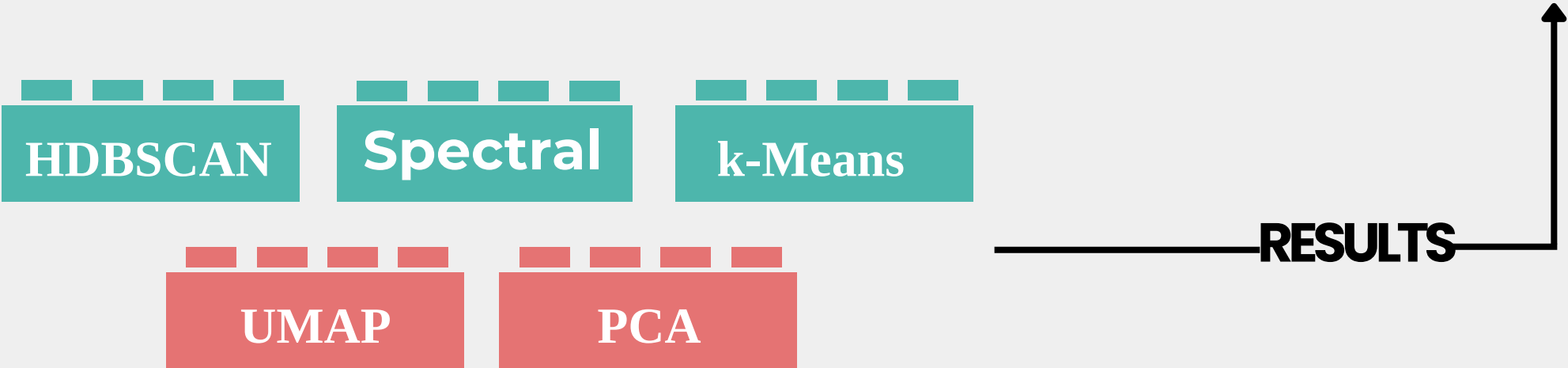
Topic Diversity Score: 0.8071



BERTopic



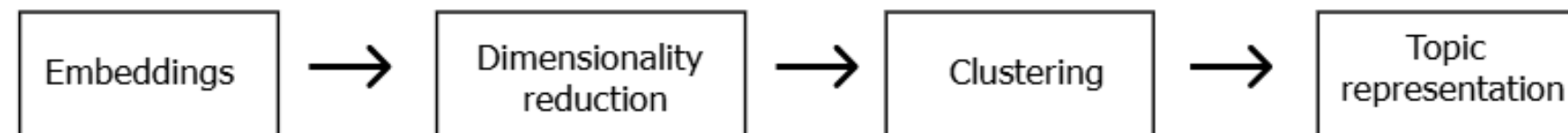
Dim. Reduction	Clustering	Silhouette	Topic Coherence	Topic Diversity
UMAP	HDBSCAN	0.0385	0.5393	0.8071
Kernel PCA	Spectral	-0.016	0.6919	0.5200
Kernel PCA	K-Means	0.0345	0.5691	0.5267
UMAP	Spectral	0.0198	0.5470	0.5733
UMAP	K-Means	0.0147	0.05681	0.5400



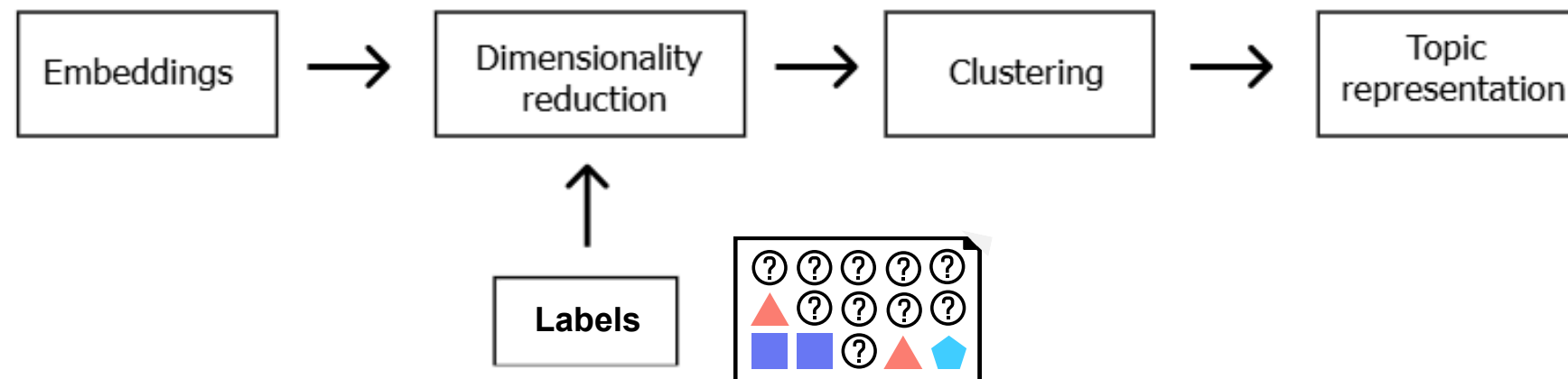
[Results computed using the default approach and hyperparamters, without stopwords removal.]

Topic Modeling: **THREE APPROACHES COMPARED**

TRADITIONAL



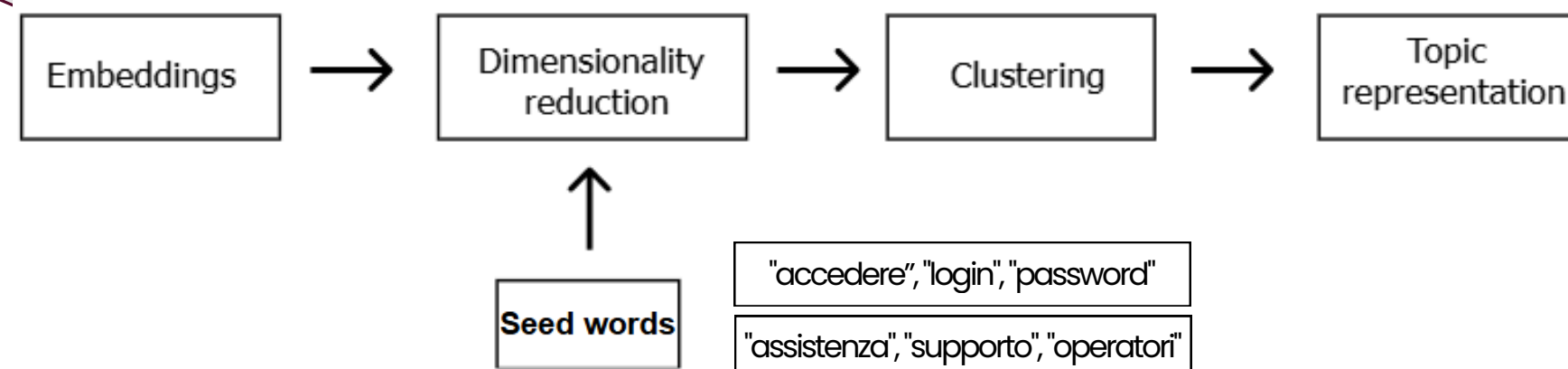
SEMI-SUPERVISED



A subset of **15%** of the reviews, **labeled** using the company's system, is used as supervision.

→ The labels provides partial supervision by guiding the UMAP embedding space to bring documents with the same label **closer**

GUIDED

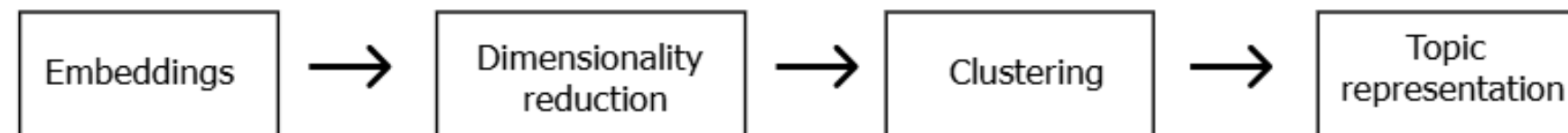


Manually defined **seed topics** are provided to BERTopic to assign reviews to the most similar predefined topic.

→ During **c-TF-IDF** phase, seed words are upweighted to increase their prominence in topic representations.

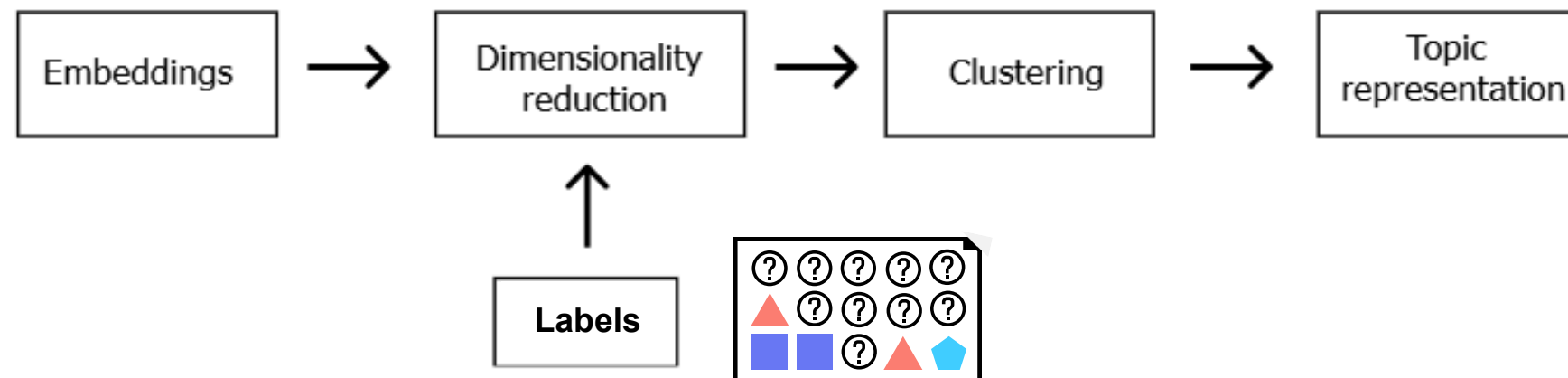
Topic Modeling: **THREE APPROACHES COMPARED**

TRADITIONAL

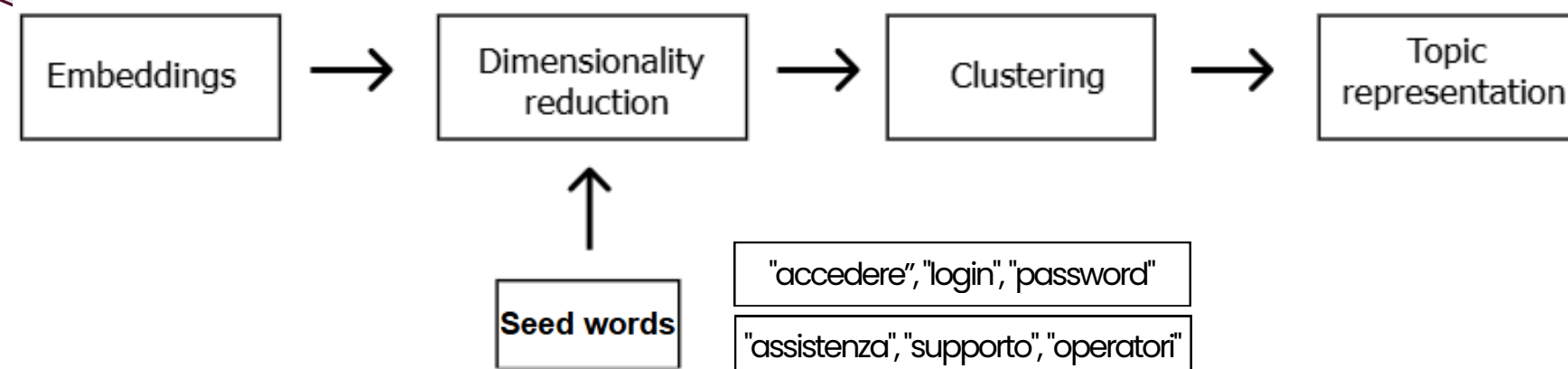


Silhouette	Topic Coherence	Topic Diversity
0.0636	0.5375	0.8071

SEMI-SUPERVISED



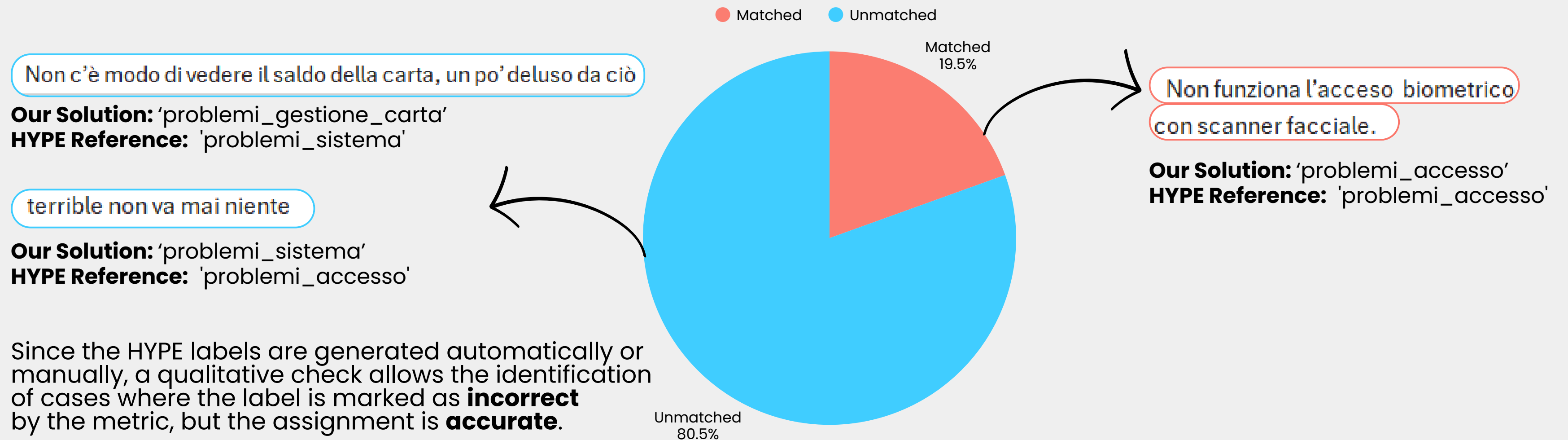
GUIDED



[Results computed using the default architecture based on UMAP and DBSCAN, without stopwords removal.]

Topic Taxonomy ALIGNMENT RESULTS

Once taxonomy alignment is performed, alignment quality is assessed using a **document-level precision metric**, computed by measuring the overlap between the **inferred taxonomy labels** and the **reference labels** for each review.



Since the HYPE labels are generated automatically or manually, a qualitative check allows the identification of cases where the label is marked as **incorrect** by the metric, but the assignment is **accurate**.

[Results computed with the single-label assignment using the best configuration, according to the evaluation metrics.]

EXTRA TASKS

Multi-Label Classification

Some reviews cover **more than one** issue.

BERTopic

For each review, we use BERTopic's **topic probability distribution**.

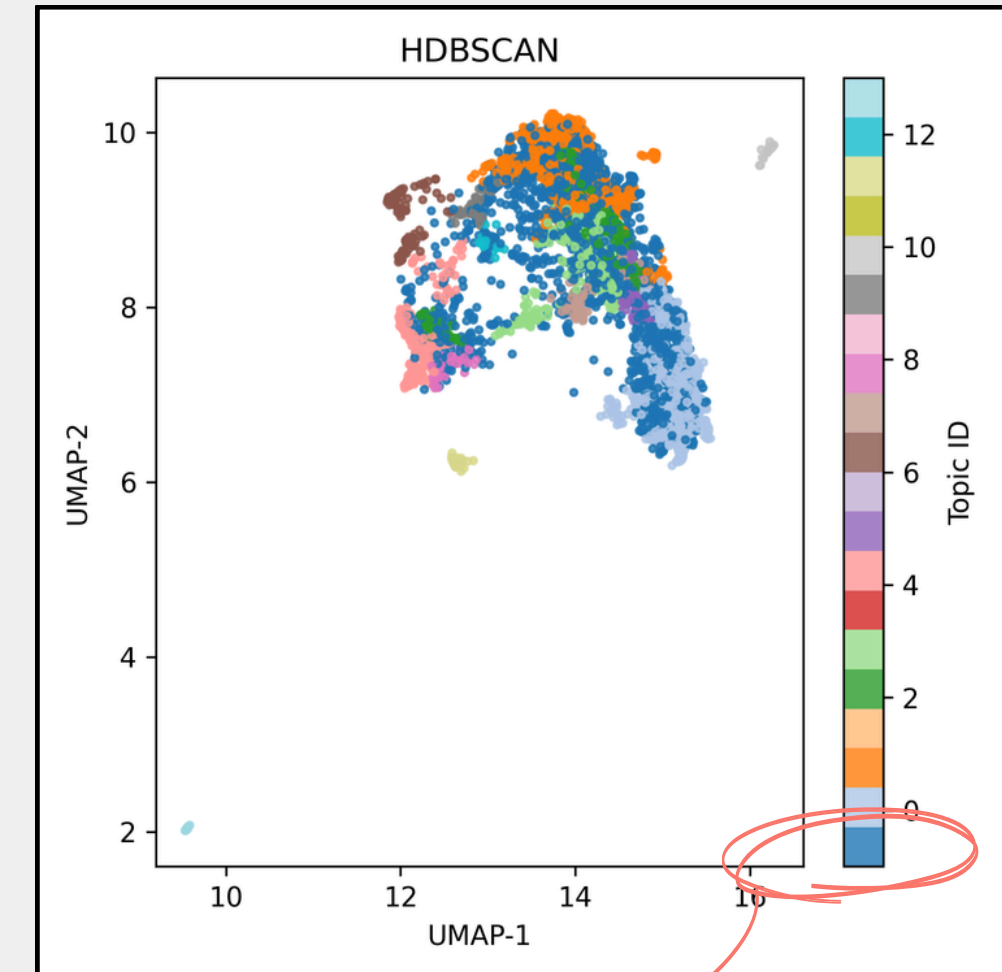
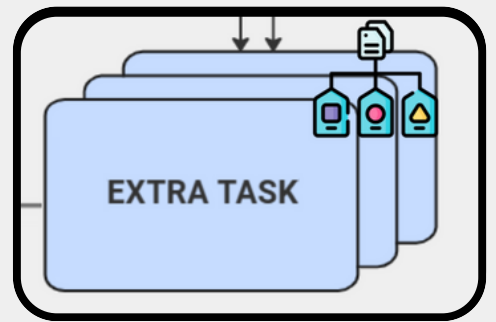
Up to three topics are assigned when their probability is $\geq 85\%$ of the top score.

Più si aggiorna e più da problemi poi contac center praticamente inesistente.

App Update Issues: **P= 0.1301**

Support Issues: **P= 0.1178**

The initial topic modeling output produced a high number of **outliers**, so the same approach was applied to reduce them.



OUTLIERS

Outlier
[Topic: -1]

Any topic
score $> 20\%$?



Reassign the review
to that topic



Keep the review
as outlier

BEST CONFIGURATION



Architecture: UMAP + HDBSCAN

Stop-word strategy: None

Topic modeling approach: Traditional

Number of topics: Fixed to 14

Outliers reduction

Silhouette	Topic Coherence	Topic Diversity	Outliers
------------	-----------------	-----------------	----------

0.0636	0.5375	0.8071	19.12%
--------	--------	--------	--------

UMAP (n_neighbors = 20, n_components = 5, min_dist = 0.0, metric = "cosine")
HDBSCAN (min_cluster_size = 20, min_samples = 5)

0_accedere_aggiornamento_riesco_funziona

1_carta_soldi_assistenza_dopo

2_password_accedere_cambiare_riesco

3_video_registrazione_aggiornamento_fa

4_operatore_chat_parlare_bot

5_fa_bonifici_funziona_mai

6_accesso_biometrico_face_id_valido

7_lenta_molto_lentezza_troppo

8_notifiche_pubblicità_spam_promozioni

9_assistenza_servizio_pessima_assistenza_inesistente

10_carta_ricaricare_nuova_altra

11_bitcoin_vendere_portafoglio_connessione

12_questionario_dato_do_tolgo

13_numero_opzione_cambiato_logicamente

LLM

Problemi con l'accesso biometrico che non riesce ad autenticarsi correttamente.

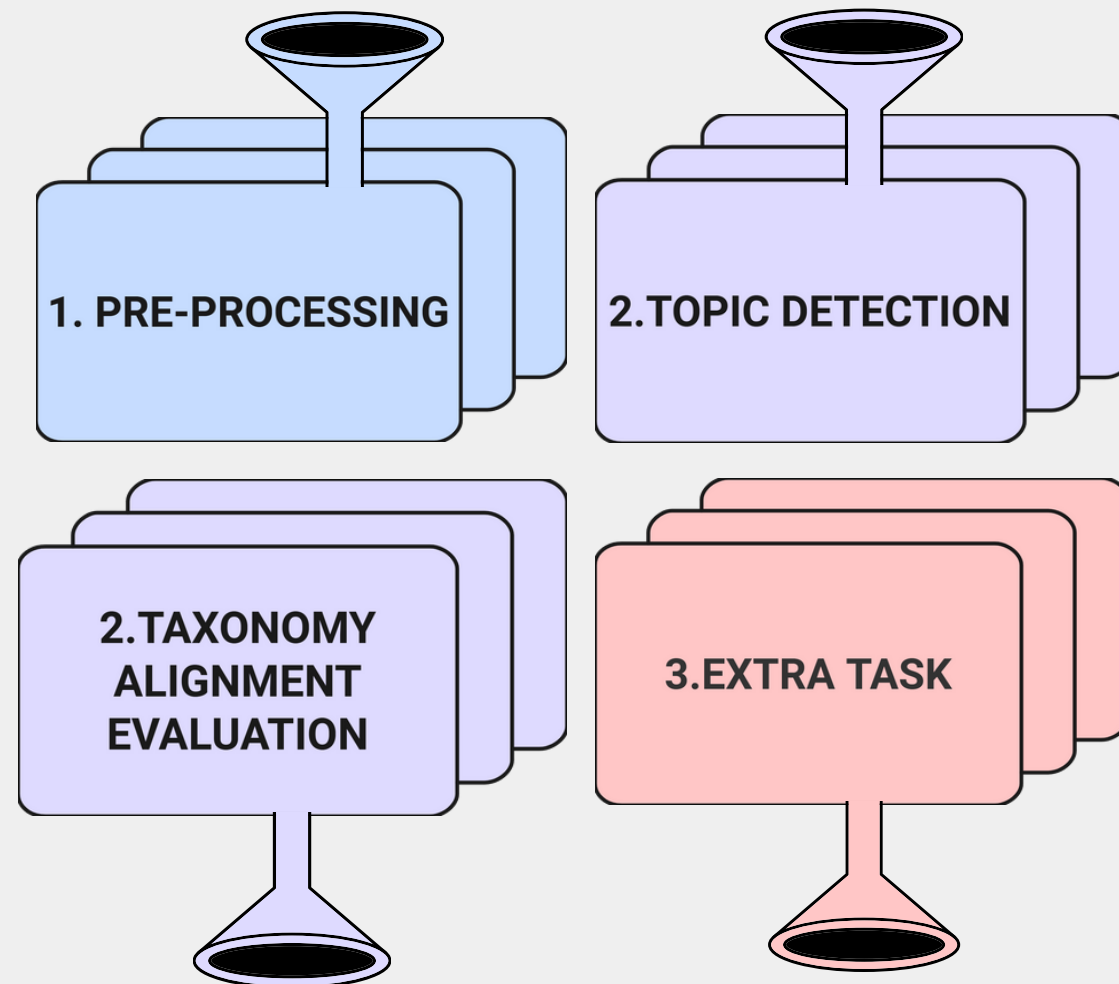
Problemi relativi alle notifiche push che includono pubblicità e non sono pertinenti.

Problemi relativi alla gestione e trasferimento dei fondi (bitcoin) nel portafoglio.

Problemi con il servizio clienti che non risponde alle richieste degli utenti.

SUMMARY OF KEY POINTS

Standard cleaning, neutral review reclassification, **stopword strategy** evaluation and **machine translation** impact analysis.

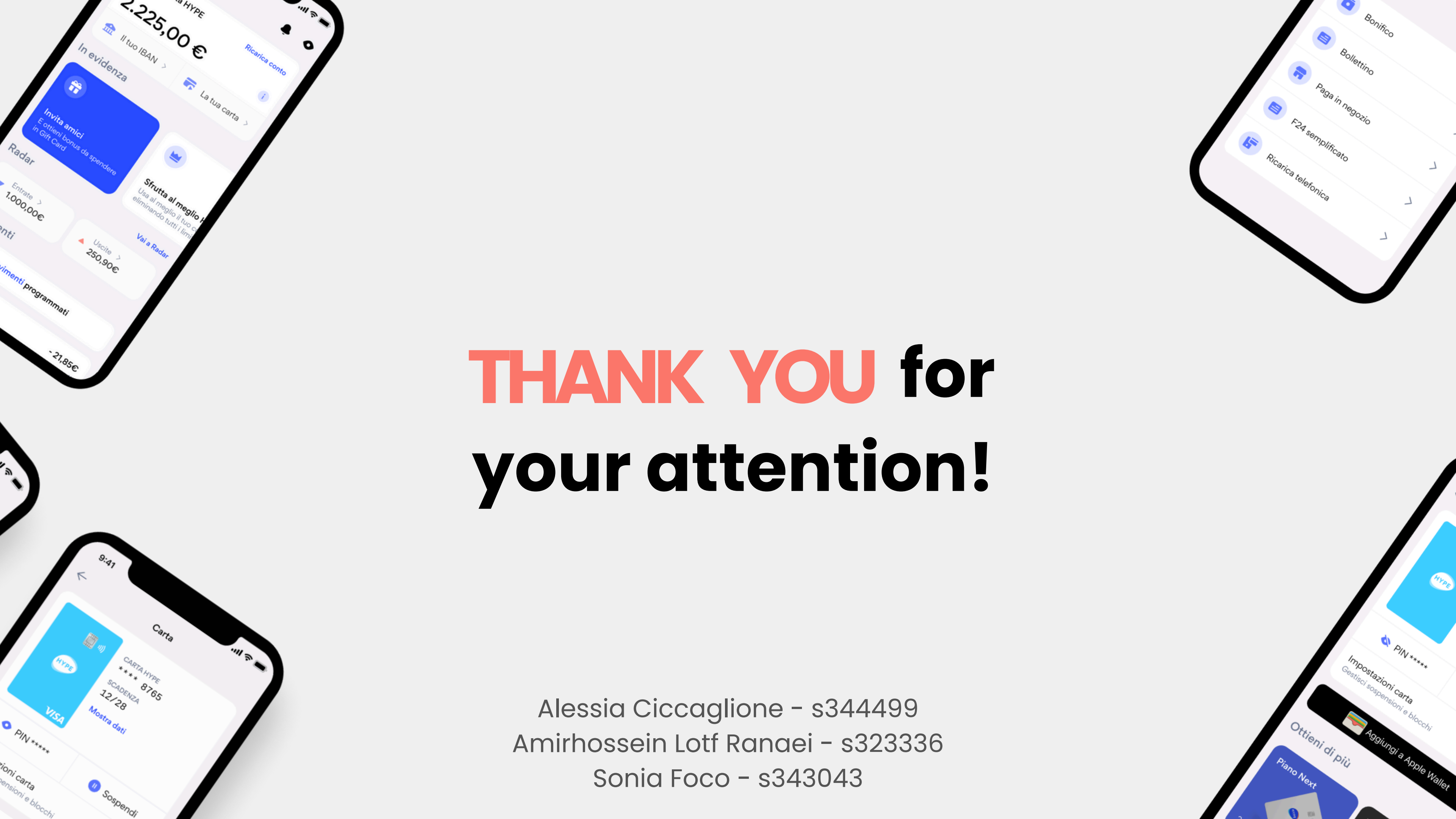


Discovered topics were mapped to the HYPE taxonomy using **embedding-based cosine similarity** and evaluated through **precision** and **qualitative** inspection.

BERTopic was compared with **LDA** and **NMF** across multiple **architectures** and modeling **strategies**.

Multi-label classification was also used for **outlier** reduction.



The background features a collage of four smartphone screens displaying the HYPE app interface. The top-left screen shows the account balance of 2.225,00 € and various service buttons like 'Invita amici', 'Radar', and 'Sfrutta al meglio'. The top-right screen displays a menu with options such as 'Bonifico', 'Bollettino', 'Paga in negozio', 'F24 semplificato', and 'Ricarica telefonica'. The bottom-left screen shows a digital representation of a Visa HYPE card with its details and a PIN. The bottom-right screen shows the 'Impostazioni carta' (Card Settings) section, including options for PIN, card suspension, and adding the card to Apple Wallet.

THANK YOU for your attention!

Alessia Ciccaglione - s344499
Amirhossein Lotf Ranaei - s323336
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